FACULTY OF COMPUTING

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(SECP3213) BUSINESS INTELLIGENCE

SECTION 02

GROUP BIII

PROJECT:

IMPROVING U.S. AIRLINE OPERATIONS THROUGH FLIGHT DELAY ANALYSIS

PHASE 1

LECTURER: DR. NOORFA HASZLINNA BINTI MUSTAFFA

STUDENT NAME	MATRIC NO
GUI KAH SIN	A23CS0080
SABRINA HENG WEI QI	A23CS0265
WOO CHENG SHUAN	A23CS0283

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1.0 Problem Background

1.1 Business Context

The U.S. commercial aviation industry is a complex and massive transportation system serving hundreds of millions of passengers annually. Operational efficiency is crucial to the success of any airline, airport, or regulatory body (such as the Federal Aviation Administration (FAA)), as they rely on operational efficiency to ensure profitability, safety measures and customer satisfaction. However, flight delays and cancellations remain a frequent and costly problem for the aviation industry.

In 2023, the U.S. aviation industry continued its recovery from the post-pandemic crises, but also faced numerous challenges, such as staff shortages, an aging fleet, unpredictable weather and ever increasing passenger demand. These operational challenges were not faced by all airlines which some consistently performed poorly in terms of on-time performance and certain aircraft types, routes and airports were more strongly correlated with flight delays and cancellations. While Bureau of Transportation Statistics (BTS) provides extensive operational data, many airlines still lack a unified, insight-driven business intelligence (BI) system to integrate flight performance, cancellation information and aircraft information into a unified database, thus failing to inform strategic decision-making.

1.2 Importance of the Problem

Flight delays and cancellations have a significant impact on multiple stakeholders including operational disruptions directly affect an airline's finances. For example, crew overtime pay, aircraft relocation costs, compensation payments to passengers and reputational damage.

Passengers become impatient if wait times are too long and prolonged delays can even attract regulatory attention. Passenger travel disruptions can trigger a cascading effect, leading to missed flights, unplanned accommodation costs and lost productivity, especially during peak seasons such as summer and year-end holidays.

Airport operators and regulators need a deep understanding of delays at the airport and route levels to adjust airport capacity, allocate gates and manage air traffic. This reminds us that system failures can have a ripple effect across the entire system. For example, the FAA's NOTAM (Notify-of-Flight) system failure in January 2023 affected thousands of flights nationwide.

Without a unified analytical framework, stakeholders are left dealing with fragmented and siloed data and information. Business intelligence (BI) solutions that integrate flight, cancellation and aircraft data can provide a solid foundation for shifting from reactive crisis management to proactive performance optimization.

1.3 Expected Impact

The purpose of developing this BI solution is to achieve the following impacts:

Operational Improvement: Airlines can identify which routes, carriers and aircraft types are most prone to delays, enabling them to implement strategies such as aircraft redesign and route optimization.

Cost Reduction: Early understanding of high-risk cancellation patterns help airlines better allocate resources and reduce unnecessary operational disruption costs, especially by differentiating between cancellations caused by carriers and those caused by weather.

Enhanced Passenger Experience: Analyzing passenger delay times and the worst-performing routes helps guide proactive passenger communication and dynamic passenger rebooking systems.

Regulatory and Safety Value: Aircraft cancellation rate trends by age and/or type help guide fleet renewal and FAA regulatory activities.

Strategic Benchmarking: Airlines can conduct self-benchmarking analysis across various aspects of the aviation industry, such as on-time performance, cancellation rates, causes of delays to identify competitive gaps and understand industry best practices.

2.0 Objectives

- To create a single, analysis-ready dataset by utilizing Alteryx Designer to combine and clean the three 2023 US civil flight datasets using the shared key Tail_Number.
- To determine the main reasons and trends of flight delays across airlines, routes, airports and time periods, by using US flight operations data from 2023.
- To investigate the connection between delay frequency and cancellation rate and aircraft attribute including type, manufacturer and age.
- To use Tableau or Power BI to create interactive storyboards and dashboards that provide critical performance metrics for flight delay and cancellation performance.
- To offer airlines and airport operators practical business suggestions to minimize flight delays and boost operational effectiveness, based on the analysis’s conclusions.

3.0 Datasets Description

The datasets used in this project were obtained from Kaggle – 2023 US Civil Flights Delay, Meteo and Aircraft Dataset. The original dataset consists of five CSV files related to US civil flight operations, delays, weather and aircraft information. However, only three datasets were selected as they are the most relevant to the project topic, Improving U.S. Airline Operations Through Flight Delay Analysis.

3.1 US Flights 2023 Dataset

3.1.1 Source of Dataset

The US Flights 2023 Dataset was obtained from Kaggle. This dataset contains detailed US operational records for the year 2023, excluding cancelled and diverted flights. The dataset includes 24 of columns and it is suitable for analysing flight performance, operational efficiency and delay trends in the aviation industry.

Dataset Link: https://www.kaggle.com/datasets/bordanova/2023-us-civil-flights-delay-meteo-and-aircraft?resource=download&select=US_flights_2023.csv

3.1.2 Type of Data

The US Flights 2023 Dataset is a structured dataset stored in CSV format. The dataset contains numerical, categorical and date-based variables related to flight schedules, date, delays and airline carriers.

3.1.3 Key Variables/Features

Variable	Data Type	Description	Example Value
FlightDate	Date	Date the flight operated	2023-01-02
Airline	String	Name of the airline carrier	Southwest Airline Co.
Tail_Number	String	Unique aircraft registration identifier	N605LR N331PQ
Dep_Airport	String	Departure airport code	ATL
DepTime_label	String	Time of day category (morning/afternoon/evening)	Morning Afternoon
Dep_Delay	Fixed Decimal	Departure delay in minutes (negative = early)	-3 35
Distance_Type	String	Flight distance category (short/medium/long haul)	Short Haul >1500Mi Medium Haul <3000Mi
Arr_Airport	String	Arrival airport code	LGA
Arr_Delay	Integer	Arrival delay in minutes (negative = early)	-12 40
Aircraft_age	Float	Age of the aircraft in years since manufacture	16 4

3.2 Flight Cancelled and Diverted Dataset

3.2.1 Source of Dataset

The Customer Personality Analysis Dataset was acquired from Kaggle. It contains records of cancelled and diverted flights in the United States during 2023. It consists of 23 columns related to flight operations, airline information, departure and arrival airports and flight disruption details. The dataset is suitable for analysing cancellation patterns, diversion trends, delay causes and operational disruptions in airline operations.

Dataset link:

https://www.kaggle.com/datasets/bordanova/2023-us-civil-flights-delay-meteo-and-aircraft?resource=download&select=Cancelled_Diverted_2023.csv

3.2.2 Type of Data

The Flight Cancelled and Diverted Dataset is a structured dataset stored in CSV format. The data is organized into rows and columns, allowing efficient preprocessing and analysis using Business Intelligence tools such as Alteryx Designer and Power BI. The dataset contains numerical, categorical and date-based variables related to flight delays, cancellations, diversions and airline performance.

3.2.3 Key Variables/Features

Variable	Data Type	Description	Example Value
FlightDate	Date	Date the flight operated	2023-01-02
Airline	String	Name of the airline carrier	Endeavor Air Southwest Airline Co.
Tail_Number	String	Unique aircraft registration identifier	N605LR N331PQ
Cancelled	Boolean	Whether the flight was cancelled (1 = yes, 0 = no)	0 / 1
Diverted	Boolean	Whether the flight was rerouted to an unplanned airport (1 = yes, 0 = no)	0 / 1
Dep_Airport	String	Departure airport code	BDL ATL
DepTime_label	String	Time of day category (morning/afternoon/evening)	Morning Afternoon
Distance_Type	String	Flight distance category (short/medium/long haul)	Short Haul >1500Mi Medium Haul <3000Mi
Arr_Airport	String	Arrival airport code	LGA ATL

3.3 US Flight January 2024 Dataset

3.3.1 Source of Dataset

The Inventory Forecasting Dataset was obtained from Kaggle. This dataset contains updated US flight operational records for January 2024, including information related to airline operations, departure and arrival airports, flight delays and aircraft details but excluding cancelled and diverted flights. The columns are same as with US Flight Dataset 2023 consists of 24 columns.

Dataset Link: https://www.kaggle.com/datasets/bordanova/2023-us-civil-flights-delay-meteo-and-aircraft?resource=download&select=maj+us+flight+-+january+2024.csv

3.3.2 Type of Data

The US Flight January 2024 Dataset is a structured dataset stored in CSV format. The dataset is organized into rows and columns and contains numerical, categorical and date-based variables related to flight schedules, delay categories, airport operations and aircraft information.

3.3.3 Key Variables/Features

Variable	Data Type	Description	Example Value
FlightDate	Date	Date the flight operated	2023-01-02
Airline	String	Name of the airline carrier	Endeavor Air Southwest Airline Co.
Tail_Number	String	Unique aircraft registration identifier	N605LR N331PQ
Dep_Airport	String	Departure airport code	BDL
DepTime_label	String	Time of day category (morning/afternoon/evening)	Morning Afternoon
Dep_Delay	Fixed Decimal	Departure delay in minutes (negative = early)	-3 35
Distance_Type	String	Flight distance category (short/medium/long haul)	Short Haul >1500Mi Medium Haul <3000Mi
Arr_Airport	String	Arrival airport code	ATL
Arr_Delay	Integer	Arrival delay in minutes (negative = early)	-12 40
Aircraft_age	Float	Age of the aircraft in years since manufacture	16 4

4.0 Proposed BI Approach

This project will adopt a Business Intelligence (BI) approach by using Alteryx Designer for data preparation and Power BI or Tableau for data visualization and dashboard development. The main objective of this approach is to transform large-scale aviation datasets into meaningful and actionable insights that can support airline operators, airport authorities and aviation regulators in understanding the factors contributing to flight delays and cancellations in the United States during 2023.

The project integrates three aviation-related datasets, namely **US_flight_2023.csv**, **Cancelled_Diverted_2023.csv** and **maj us flight.csv**, to provide a more comprehensive view of airline operational performance. Through the application of BI tools, the project aims to improve the understanding of delay patterns, cancellation causes and aircraft-related operational performance in a structured and data-driven manner.

4.1 Use of Alteryx and Visualization Tools

In this project, Alteryx Designer been chosen as the main platform to complete the data preparation, data integration and data preprocessing. Since the datasets comes from different files and having different types of operational information, Alteryx play an important role in organizing and preparing the data before conducting analysis.

The first step is to import the three datasets into Alteryx by using the Input Data Tool. The US_flight_2023.csv dataset is the primary operational file with flight schedules, timing performance and delay information. The Cancelled_Diverted_2023.csv dataset provides information regarding cancelled and diverted flights, while the maj_us_flight.csv dataset acts as a reference table containing aircraft-related information such as manufacturer, aircraft type, model and year of manufacture.

Once the preprocessing stage is done, the final integrated dataset will be exported to PowerBI or Tableau for visualization. These visualization platforms will be leveraged to create interactive dashboards and storyboards that enable users to dynamically explore operational performance. Interactive filters and drill down will allow users to explore performance by airline carrier, airport, route, aircraft type and time period.

The project will integrate Alteryx with visualization tools to enable a more efficient approach to aviation operational performance analysis than traditional analysis.

4.2 Data Integration, Data Cleaning and Data Transformation Methods

This project will require data integration, which will entail merging the three datasets together based on the common key attributes, Tail_Number, which is unique to an aircraft. The aircraft operational data will be joined with the aircraft cancellation records and aircraft reference data using the Join Tool in Alteryx. This integration process allows flight delay data, cancellation status and aircraft attributes to be analysed in a single dataset.

Several data preprocessing steps will be performed to enhance data quality and consistency for data cleaning. If any important variables are missing, such as delay duration, cancellation reasons and aircraft details, this will be identified and dealt with appropriately. Duplicate records will also be checked and removed to avoid inaccurate analysis results. Moreover, the data format will be standardized to make it compatible across the datasets, especially for time-related variables and categorical fields, where data formats are inconsistent.

Regarding data transformation, some processes will be used to make the data more appropriate to be analyzed. The relevent variables will be chosen and the irrelevant attributes will be eliminated to make the workflow efficient. You can also use the Formula Tool to create additional calculated fields, including:

- Total Delay Duration – Sums up various categories of delay.
- Aircraft Age based on year of manufacture
- Delay Severity Category which classifies delays as low, medium or high severity
- To streamline the classification of disrupted flights, Cancelation Status Indicator was added

Aggregation techniques may also be applied to summarize delay frequency, cancellation rate and operational performance across airlines, airports, routes and aircraft types.

4.3 Proposed Analysis

The project will involve analysing flight delay, cancellation and operational performance differences between airlines, routes, airports and aircraft types in the United States in 2023. The analysis aims to give a better understanding of operational inefficiencies and to determine the main causes of flight disruptions.

Firstly, the performance analysis of airlines will be done, which will compare the average duration of delay among different airline companies. This analysis is designed to discover airlines that perform better or worse in terms of punctuality and delay management. This comparison allows stakeholders to assess the consistency of the service quality provided by the airlines and the possible gaps in performance.

Second, cause analysis will be delayed to examine the main factors that contribute to flight delays. The various categories of delays such as carrier delay, weather delay, National Aviation System (NAS) delay, security delay and late aircraft delay will be analyzed to identify the most significant factors causing operational disruptions. This analysis may help airlines and airport operators implement more targeted mitigation strategies.

Third, temporal trend analysis will be performed to investigate the delay patterns over time. Delays will be analysed on a monthly and seasonal basis across the year 2023 to determine when there are greater delays in operations. This analysis is valuable to understand if there are more delay occurrences during peak travel times, during holidays or certain operational times.

Fourth, route and airport performance analysis will be carried out to pinpoint airports and routes that are affected by greater delays and disruptions. Geographic performance analysis can identify congestion hot spots or operational bottlenecks at certain locations.

Finally, cancellation and diversion analysis will be conducted to explore the cancellation and diversion flight patterns by airline. The goal of the analysis is to identify the most vulnerable airlines from the disruption perspective, and to pinpoint the top drivers of cancellation events.

The project will provide practical lessons that can benefit airlines, airport operators and aviation regulators to enhance the efficiency and minimize disruptions in operations.

4.4 Proposed Visual Forms

To support the analysis and present meaningful insights effectively, the following visual forms are proposed:

1. Bar Chart – Average Departure Delay by Airline
   A bar chart will be used to compare the average departure delay of different airline carriers. This visualization helps stakeholders to easily recognize the airlines with the best and worst performance in terms of delays and helps to benchmark the aviation industry.

2. Stacked Bar Chart – Delay Cause Breakdown per Airline
   The distribution of delay causes will be displayed in a stacked bar chart by airline, comprising of the following delay causes: carrier delay, weather delay, NAS delay, security delay and late aircraft delay. This visual will show the primary reasons for delays for each airline and identify operational weaknesses.

3. Line Chart – Monthly Delay Trend
   The average departure delay trend for each month will be analysed using a line chart. This visualization enables stakeholders to see seasonal trends, travel disruptions at peak times and longer-term trends in operations over time.

4. Geographic Map – Route Delay Heatmap by Airport
   The delay intensity will be mapped at the geographic level for departure and arrival airports in the United States. This visual is used to identify geographic hotspots of flight disruption and to gain a better understanding of the operational performance at the airport level.

5. Treemap/ KPI Card – On-Time Performance (OTP%) by Airline
   The percentages of on-time performance will be displayed using a treemap or KPI card visualization by airline carrier. Larger sections or indicators will indicate more successful airlines and enable stakeholders to easily compare the level of punctuality of airlines.

These visual forms are chosen to offer a clear and interactive view of airline operational performance from various angles, such as delays, cancellations, route efficiency, and airline benchmarking. The dashboard seeks to facilitate more informed and evidence-based operational decision making by integrating trend analysis, geographic visualization and performance indicators.